

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

MAY EXAMINATION

UCCD2303 DATABASE TECHNOLOGY /
UCCD2203 DATABASE SYSTEMS

TUESDAY, 16 MAY 2023

TIME : 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
DIGITAL ECONOMY TECHNOLOGY

Instructions to Candidates:

This question paper consists of **Section A** and **Section B**.

Section A consists of **THREE (3)** questions. Candidates are required to **answer ALL the questions**.

Section B consists of **TWO (2)** questions. Candidates are required to **answer only ONE (1) question**. Should a candidate answer more than **ONE (1)** question in Section B, marks will only be awarded for the **FIRST** questions in the order the candidate submits the answers.

Candidates are not allowed to use a calculator.

Answer questions only in the answer booklet provided.

Section A: Answer ALL THREE (3) questions. [Total: 75 Marks]

- Q1. (a) (i) Compare PRIMARY KEY and FOREIGN KEY in terms of the supporting of the null values, record uniqueness and the referencing method with other relations. (6 marks)
- (ii) Based on the answers in Q1(a)(i), provide any **ONE (1)** example of relational scheme and identify the PRIMARY KEY and FOREIGN KEY. (2 marks)
- (b) Database views are widely used as a great way for database optimization defined by a query. Argue **THREE (3)** reasons why base tables are still preferred over views under certain situation(s). (6 marks)
- (c) ABC Sdn. Bhd. would like to keep track of its computer system components. They want to keep the information such as systemName, ipAddress, location, computer type, computer manufacturer, computer model and computer serial numbers for each system and its components.
- (i) Determine whether the scenario above is an aggregation or a composition. (2 marks)
- (ii) Draw **ONE (1)** Extended Entity Relationship Diagram (EERD) based on the given scenario. (4 marks)
- (d) Null value indicates an absent value and could not be avoided in some cases. Argue how the existence of null value could create **TWO (2)** potential problems in the database transactions. (5 marks)
- [Total : 25 marks]

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Q2. (a) There are various types of database failures that can affect database processing which might be affecting the main memory while others might involve secondary storage.

(i) Identify any **TWO (2)** types of database failure. (4 marks)

(ii) Suggest any **TWO (2)** database recovery methods used in relational databases. (4 marks)

(b) Given a scenario:

In the CUSTOMER table, transaction T1 holds a lock on some records and needs to update some records in the SALES table. Simultaneously, transaction T2 holds locks on some records in the SALES table and needs to update the records in the CUSTOMER table held by Transaction T1.

Figure 2.1

(i) Identify the **TWO (2)** problems based on the given scenario in Figure 2.1. (4 marks)

(ii) Determine the state of transactions in Figure 2.1. Justify your answers. (4 marks)

(iii) Suggest any **THREE (3)** general techniques for handling the problems based on the answers in Q2(b)(i). (3 marks)

(c) Transaction recovery is the process of recovering the failed transaction where it is removed from the database failures. Differentiate between DEFERRED UPDATE and IMMEDIATE UPDATE recovery techniques. (6 marks)

[Total : 25 marks]

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- Q3. Figure 3.1 shows a fragment of an Extended-Entity-Relationship model describing customer orders. Refer to Figure 3.1 and answer all the questions below.

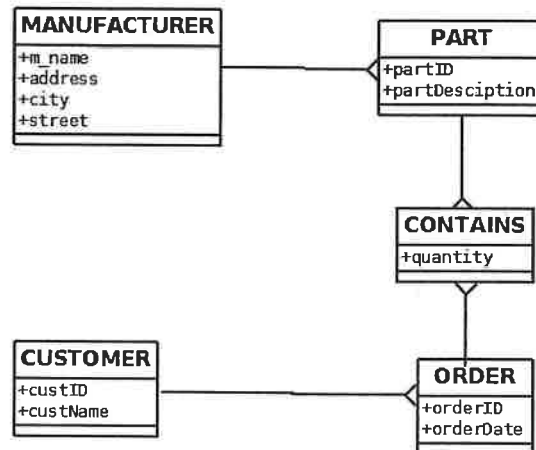


Figure 3.1

- (a) (i) Identify the ENTITIES and clearly indicate all primary and foreign keys (also state into which relation they are the keys). (9 marks)
- (ii) Based on Figure 3.1, suggest **ONE (1)** example of COMPOSITE ATTRIBUTE and **ONE (1)** example of MULTI-VALUED ATTRIBUTE. (4 marks)
- (b) Relate the **FOUR (4)** basic properties of a transaction with **ONE (1)** real-life example for each. (8 marks)
- (c) Refer to the following relations, answer all the questions below:

SUPPLIERS(sID, sName, address)
 PARTS(pID, pName, colour)
 CATALOG(sID, pID, price)

Figure 3.2

- (i) Write **ONE (1)** relational algebra to display all the part names which are in red color. (2 marks)
- (ii) Write **ONE (1)** relational algebra to display all the prices for parts that are red or green. (2 marks)

[Total : 25 marks]

UCCD2303 DATABASE TECHNOLOGY / UCCD2203 DATABASE SYSTEMS**Section B [Total: 25 marks]**

This section consists of TWO (2) questions. Answer ONLY ONE (1) question in Section B.

- Q4. (a) XYZRetail Sdn. Bhd. is considering to migrate their existing database system into a distributed environment by using a Distributed Database Management Systems (DDBMS). There are DDBMS that are handling the actual transactions against your checking/saving accounts, the processing of a credit or debit-card transaction, other systems that deal with mortgages/loans, yet they are all operating on the same account. As an IT consultant, you are required to explain to the management about the functionalities and components of the new system.
- (i) Recommend any **FOUR (4)** major DDBMS components which suit the needs of XYZRetail Sdn. Bhd. (8 marks)
 - (ii) Explain any **SIX (6)** functionalities which distinguish DDBMS from DBMS. (6 marks)
 - (iii) Predict any **THREE (3)** concurrency control issues in the multi-user environment of XYZRetail Sdn. Bhd. (3 marks)
- (b) Explain any **TWO (2)** differences between SQL and NoSQL. Provide **ONE (1)** real-life example for each. (6 marks)
- (c) Outline any **TWO (2)** issues with Mobile DBMSs. (2 marks)
- [Total : 25 marks]

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- Q5. (a) In the context of transaction processing of Relational Database Management Systems (RDBMS), the ACID transactions have become a bottleneck in scaling RDBMS to handle big transactional data especially with the evolution of mobile devices. Argue why ACID transactions are not suitable for mobile applications in the present era. (8 marks)
- (b) A cloud database is a database service built and accessed through a cloud platform whereby serving most of the functions as a traditional database. Name **THREE (3)** Cloud Implementation Types. (3 marks)
- (c) Hadoop is a part of Apache project. It emerged as a solution to the Big Data challenges for distributed storage and distributed processing of large data sets. Elaborate the Hadoop Ecosystem which allow the system to continue operating in case of node failure. (6 marks)
- (d) Compare a classical database application with a mobile database application in terms of "structure of data object", "structure of transaction", "size of data object" and the "size of data sets". (8 marks)
- [Total : 25 marks]
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